

Dwella

Dwella is a comprehensive digital platform designed to bridge communication and management gaps for rental properties. It provides a unified space for tenants, landlords, and administrators to efficiently manage properties, leases, payments, maintenance, and communication. The system emphasizes role-based access, real-time interactions, and robust financial tracking.

✨ Features

🔑 Core Features

- **User Authentication:** Secure registration and login using JWT tokens, with password hashing via bcrypt or Argon2.
- **Role-Based Access Control (RBAC):**
 - **Tenants:** View lease details, pay rent online, submit & track maintenance requests, message landlord, access payment receipts.
 - **Landlords:** Manage properties & units, invite tenants, view payment history, create maintenance tasks, send announcements, view financial reports, manage leases.
 - **Admins:** Oversee all accounts, handle disputes, manage platform settings, view application logs.
- **Maintenance Tracking:** Centralized ticketing system with statuses like `Draft`, `Open`, `In Progress`, `Resolved`, etc.
- **Real-Time Chat:** WebSocket-based messaging between landlords and tenants. All occupants of a property belong to a "group" and can create private chats within the same group.

🏠 Property & Unit Management

- Landlords can add, edit, or remove properties (address, photos, amenities).
- Each property can contain multiple units (apartments, rooms) with individual lease terms.
- Unit-level details: rent amount, deposit, occupancy status, square footage, etc.

📄 Lease Management

- Create digital lease agreements using customizable templates.
- Built-in e-signature support (native or via DocuSign).
- Automated reminders for lease renewal (60/30/7 days before expiry).
- Secure storage of all lease documents.
- Lease lifecycle tracking: `draft` → `active` → `expired` → `renewed`.

💳 Rent Payments

- Integration with payment gateways (Stripe, PayPal, bank transfer).

- Recurring payments (auto-pay) for tenants.
- Automatic ledger updates and receipt generation.
- Late fee calculation and notifications.
- Support for partial payments and custom payment plans.

Maintenance Requests

- Tenants submit requests with category (plumbing, electrical, etc.), description, and photos.
- Landlords assign priority and status (open, in progress, resolved).
- Optional integration with third-party contractor services to assign jobs.
- Tenants receive updates via in-app notifications and email.

Communication Hub

- Threaded messaging between landlord and tenant (per property or per unit).
- Announcements: broadcast messages to all tenants of a property.
- Notification center supporting email, SMS, and push notifications.
- Centralized inbox for all communications.

Financial Dashboard (Landlord)

- Income/expense tracking per property.
- Monthly/yearly profit & loss statements.
- Exportable CSV/PDF reports.
- Tax-ready summaries (e.g., Schedule E categories).
- Visual charts and graphs for quick insights.

Additional Features

- **Tenant Screening:** Landlords can request background/credit checks through integrated services (e.g., TransUnion SmartMove).
- **Reviews & Ratings:** Tenants can rate their rental experience; landlords can respond.
- **Automated Move-In/Move-Out Checklists:** Digital checklists with photo documentation to reduce security deposit disputes.
- **Mobile App:** React Native version for iOS and Android.
- **Accounting Integration:** Sync with QuickBooks or other accounting software.
- **Asynchronous Task Processing:** Production deployment integrates Celery/RabbitMQ with Redis broker to handle heavy tasks like mass notifications and payment processing.

Tech Stack

Layer	Technology
Frontend	React, React Router, Bootstrap 5, Zustand (state & token management), Axios
Backend	FastAPI (Python 3.9+), Pydantic, Uvicorn, WebSockets
Database	MariaDB, SQLAlchemy ORM, Alembic (migrations)
DevOps	Docker, Docker Compose, GitHub Actions (CI/CD), Nginx
Other	Celery + RabbitMQ + Redis (async tasks), Stripe/PayPal SDK, DocuSign API

API Routes

Authentication

Method	Endpoint	Description	Access
POST	/api/auth/register	Register new user	Public
POST	/api/auth/login	Login and receive JWT tokens	Public
POST	/api/auth/refresh	Refresh access token	Public
POST	/api/auth/logout	Invalidate token	Auth

Users

Method	Endpoint	Description	Access
GET	/api/users/me	Get current user profile	Auth
PUT	/api/users/me	Update own profile	Auth
DELETE	/api/users/me	Delete own account	Auth
GET	/api/users	List users (admin only)	Admin
PUT	/api/users/{id}/role	Change user role	Admin

Properties & Units

Method	Endpoint	Description	Access
GET	/api/properties	List properties (filtered by role)	Auth
POST	/api/properties	Create new property	Landlord
GET	/api/properties/{id}	Get property details	Auth
PUT	/api/properties/{id}	Update property	Landlord
DELETE	/api/properties/{id}	Delete property	Landlord
GET	/api/properties/{id}/units	List units in property	Auth
POST	/api/properties/{id}/units	Add unit to property	Landlord
PUT	/api/units/{id}	Update unit details	Landlord
DELETE	/api/units/{id}	Remove unit	Landlord

Leases

Method	Endpoint	Description	Access
GET	/api/leases	List leases (tenant sees own)	Auth
POST	/api/leases	Create new lease	Landlord
GET	/api/leases/{id}	Get lease details	Auth
PUT	/api/leases/{id}	Update lease	Landlord
DELETE	/api/leases/{id}	Terminate lease	Landlord
POST	/api/leases/{id}/sign	Initiate e-signature	Tenant
POST	/api/leases/{id}/renew	Trigger renewal reminder	Landlord

Payments

Method	Endpoint	Description	Access
GET	/api/payments	List payments (tenant: own; landlord: property)	Auth
POST	/api/payments	Initiate payment	Tenant
POST	/api/payments/{id}/confirm	Confirm webhook from gateway	System
GET	/api/payments/{id}/receipt	Download receipt	Auth
POST	/api/payments/plans	Set up payment plan (landlord)	Landlord

Maintenance

Method	Endpoint	Description	Access
GET	/api/maintenance	List tickets (filtered by role)	Auth
POST	/api/maintenance	Create new request	Tenant
GET	/api/maintenance/{id}	Get ticket details	Auth
PUT	/api/maintenance/{id}	Update status/priority	Landlord
POST	/api/maintenance/{id}/assign	Assign contractor	Landlord
POST	/api/maintenance/{id}/photos	Upload photos	Tenant

Chat & Communication

Method	Endpoint	Description	Access
WS	/ws/chat/{group_id}	WebSocket for real-time chat	Auth
GET	/api/chat/groups	List user's chat groups	Auth
POST	/api/chat/groups	Create private chat within group	Auth
GET	/api/chat/messages/{group_id}	Fetch message history	Auth
POST	/api/announcements	Send announcement to property	Landlord
GET	/api/announcements	List announcements	Auth

Financial Dashboard

Method	Endpoint	Description	Access
GET	/api/financials/summary	Income/expense summary per property	Landlord
GET	/api/financials/pnl	Profit & loss statement (monthly/yearly)	Landlord
GET	/api/financials/export	Export reports (CSV/PDF)	Landlord
POST	/api/expenses	Add expense (with receipt OCR)	Landlord

Admin

Method	Endpoint	Description	Access
GET	/api/admin/logs	View application logs	Admin
GET	/api/admin/users	List all users	Admin
PUT	/api/admin/users/{id}/status	Suspend/activate user	Admin
GET	/api/admin/stats	Platform-wide statistics	Admin

Note: All endpoints (except public auth) require a valid JWT. Role-based access is enforced via middleware. For detailed request/response schemas, refer to the auto-generated FastAPI docs at `/docs` or `/redoc`.

System Architecture

A high-level overview:

```
1  Client (React SPA / React Native)
2      |
3      ▼
4  Nginx (reverse proxy, TLS)
5      |
6      ▼
7  FastAPI (REST/WebSocket)
8      |
9      ├── SQLAlchemy ORM → MariaDB
10     ├── Redis (cache, session)
11     ├── RabbitMQ (message broker)
12     |   |
13     |   ▼
14     |   Celery Workers (async tasks)
15     |   |
16     |   ├── Email/SMS notifications
17     |   ├── Payment processing
18     |   └── Report generation
19     └── External APIs (Stripe, DocuSign, OCR, etc.)
```